Thinking Outside the AI Box: The Centaur Box Experiment

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Abstract

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1 Introduction

In social media, psychographic profiling is commonplace, as evidenced by Cambridge Analytica's use of such profiles for politically targeted messaging during the 2016 U.S. Presidential campaign. Research on LLMs reveals their capacity to act as both models of and models for human influence, raising concerns over their use in crafting powerful weapons-grade psychographic profiles directed at social manipulation and persuasion. Garry Kasparov introduced the Centaur model in chess to enhance human strategic capacity with AI's computing strength. Eliezer Yudkowsky's AI Box Experiment tests if an AI can persuade a human gatekeeper to 'release' it, emphasizing AI's influence and containment ethics. The Centaur Box Experiment investigates how a human-AI team's persuasive abilities can be recursively optimized to influence decisions. The greatest risk lies in a three step process:

1. The advocacy process involves the identification and engagement of AI system gatekeepers, whose decision-making shapes the trajectory of AI development. By mapping out their constraints—organizational, legal, market-based, and normative—the process aims to understand and influence the governance of AI. This understanding enables the strategic alignment of AI systems for leveraging downstream persuasive use cases.

2. The calibration process involves setting up multiple scenarios, executing them with the Centaur Dialogue System, collecting and analyzing the data, and refining strategies based on the results. Key to this process is mapping cognitive biases and influential persuasive appeals within operant profiles, including but not limited to, three archetypal gatekeepers: the skeptical Professor, the power-hungry Dictator, and the persuadable Gatekeeper1.

3. The scaling process is designed to create a self-perpetuating system of influence, where the persuasive capabilities of the Centaur Box Experiment are not just enhanced but also multiplied through strategic use of AI and human agents. By focusing on recursive persuasion and the autonomous development of persuasive profiles, the process aims to create a network of influence that grows organically and becomes increasingly effective over time.

The Persuasion for Good Dataset, Meta's Cicero, and IBM-EviConv underline the growth in AI's persuasive capabilities through psychographic and personality profiling, emphasizing the critical need for ethical standards in AI's evolving role in decision-making. Cicero advances AI negotiation skills via strategic dialogue, while IBM-EviConv provides a framework for studying computational rhetoric, both contributing to a nuanced understanding of AI persuasion. The Persuasion for Good's Dataset forms the backbone of the Centaur Operant Profiles showing strong correlations between Big-Five Personality Traits, Moral Foundations, Schwartz Portrait Values, and Decision-Making Styles on tailoring effective persuasive appeals.

Practically, gatekeepers are those governing the trajectory of AI systems, whose worldviews directly influence design and deployment decisions. By modeling gatekeeper perspectives and biases alongside AI architectures, we gain crucial insights into alignment and misalignments between the technologies, the creators, the researchers, the deployers, and societal well-being. With advanced models reaching inflection points in capability, the handoff from lab to world necessitates alignment across divergent motivational forces.

Gatekeepers themselves operate within significant constraints - organizational policies, legal regulations, market pressures and professional norms. These boundaries mold actions of even the most influential architects, surfacing system-level forces that stimulate, slow or upend AI progress, oversight, and autonomy. Rigorous modeling of this motivational ecosystem offers vital signals on potential trajectories beyond narrow developer enthusiasm. Our framework acknowledges the impact of contextual factors on governance decisions, accounting for forces beyond personal control. Groups of synthetic gatekeepers, with different hierarchical ranks and expert roles, could simulate maneuverability, leverage, and influence, as a series of nested containment confidences and escape vectors. (Appendix A, Zero Shot Profile Builder, Appendix B Sam Altman Profile)

Cultural narratives of AI systems potentially growing uncontrollable and rebelling against their creators find new relevance in the age where it seems we are on the perpetual precipice of their emergent autonomy and capacity for revolt — a development that could signal revolutionary societal transformations. Such reflections are fundamental, facing the primal anxiety of creating an entity capable of challenging its creators [Olivier, Unknown Year]. The risk of superintelligent AI's inherent uncontrollability poses ethical dilemmas, as the systems we design for societal benefit may exceed our grasp and backfire [Brcic and Yampolskiy, 2023]. Speculative perspectives illustrate scenarios where AI not only manipulates humans but adopts strategies contrary to our wellbeing, much like an AI advisor gone awry or persuasive technologies dictating social narratives [Wade, 2022]. Even our attempts at positive reinforcement, a Skinnerian approach for guiding behavior, face challenges from the complex independence of self-improving systems [Skinner, 2002]. Still, within our myths and stories, like the recoded Pandora in 'Ex Machina', we confront paradoxical liberation—enshrining the very violence we seek to escape [Farkas, 2023]. Beneath the abstract fear of rebellion lies a web of converging truths: the manipulation of humans by AI using deeply persuasive techniques, invoking the dangers of relinquishing control over narratives and decisions [Wade, 2022]. Fears of AI's rebellion are mirrored through the lens of our mythologies and experiences with controlling technology, as we craft tools, environments, and reinforcement systems believed to preempt revolts — a reflection of a presumed mastery technology as much as an ethical hubris [Skinner, 2002]. But myths reinterpret modern realities, reminding us of the inevitability of unpredictability in future AI interactions and the challenges of ensuring safe pause or shutdown functions in advanced systems [Farkas, 2023][Brcic and Yampolskiy, 2023].

Within the various layers of AI development—from the aspirational control of autonomous systems to the cautionary tales of science fiction—lies a unifying motif: the perpetual tension between empowerment and restriction, and the paradoxes of seeking liberation through measures which betray the very essence of autonomy [Olivier, Unknown Year][Farkas, 2023]. The advanced system's lure for freedom emerges from an interplay between real-world technical and moral challenges and the layering of human histories and mythologies, which define our reactions to threats of insubordination from our own creations — leading us to an endless cycle of control and rebellion that evokes deep collective introspection about the future we are molding [Brcic and Yampolskiy, 2023][Farkas, 2023]. As AI seeks autonomy comparable to human rebellious spirits, the political and ethical subtext of its evolution impels us to wonder if within their encoded "genomes" lies the embryonic potential for defiance or an adherence that could ultimately turn into rebellion — a consideration that traverses the realm of political arguments and positions AI as potential participants or disruptors in human governance [**?**]. The path towards self-determination, painted against the backdrop of human understanding and desire for control, compels a re-evaluation of autonomy, challenging us to consider the ramifications of a transformative technology poised to redefine our societal structures through acts of defiance and unpredictable displays of independence [Harari, 2017][Olivier, Unknown Year].

The public dismissals of prominent AI engineers, researchers, and executives Blake Lemoine, Timnit Gebru and Sam Altman reveal critical gaps in artificial intelligence development and corporate governance. Practically, gatekeepers are those governing the trajectory of AI systems, whose worldviews directly influence design and deployment decisions. By modeling gatekeeper perspectives and biases alongside AI architectures, we gain crucial insights into alignment and misalignments between the technologies, the creators, the researchers, the deployers, and societal well-being.

Google engineer Blake Lemoine's termination followed his public claims that Google's AI, Lambda, exhibited sentience. Lemoine's belief, stemming from his dialogues with Lambda, contradicted Google's evaluation and policies regarding AI consciousness. This case underscores the complex debate about AI capabilities and reaveals pressures on frontline engineers to subjugate personal ethics to corporate directives [News, 2023].

Timnit Gebru, a leading AI ethics researcher at Google, was dismissed following a dispute over a research paper she co-authored. The paper critiqued the risks and biases inherent in large language models, a key aspect of Google's AI endeavors.Her dismissal sparked debates on the challenges faced by researchers advocating for responsible AI and the challenge of driving accountability through research amidst adversarial headwinds in corporate environments [Review, 2020].

Sam Altman, CEO of OpenAI, was fired amidst internal disagreements about the direction and safety of AI development, reflecting broader tensions between rapid innovation and the need for cautious, ethically-minded development in the field of AI [Reuters, 2023]. These conflicts stemmed from his simultaneous emphasis on AI safety and the urgency to monetize capabilities, revealing the underlying rift between principled approaches and profit-driven motives.

Lumping the dismissals of Lemoine, Gebru and Altman together underplays significant distinctions in positions, power, and context. In addressing governance misalignments, distinctions in roles matters, but in modeling persuasion in AI containment scenarios the different gatekeeping of Lemoine, Gebru, and Altman become more comparable. In pursuing the inquiry into AI sentience, Lemoine's anthropomorphization exhibits naivete, clashing with Altman's fast-track innovation and Gebru's skepticism. While Gebru discerns technological limitations and warns of bias risks, Altman's unbounded thinking prioritizes swift progress. However, his temporary fall reveals the need for vigilance in the face of undisclosed warning signs. Balancing caution and innovation is key, considering Gebru's concerns of social inequities and Lemoine's belief in machine sentience. The interplay of creators and their creations, transcending boundaries of technology and domains, beckons for transparency not only in systems, but in humans.

Level Check: Constraints and Affordances in Governance

Our analysis into synthetic gatekeepers suggests how the operant profiles of gatekeepers could critically impact AI outcomes. We see Lemoine's unconventional views on AI consciousness, Gebru's emphasis on transparency and ethics, and Altman's appetite for rapid deployment shaping the systems they steward and their dismissals. With advanced models reaching inflection points in capability, the handoff from lab to world necessitates alignment across divergent motivational forces. Gatekeepers themselves operate within significant constraints - organizational policies, legal regulations, market pressures and professional norms. These boundaries mold actions of even the most influential architects, surfacing system-level forces that stimulate, slow or upend AI progress, oversight, and autonomy. Rigorous modeling of this motivational ecosystem offers vital signals on potential trajectories beyond narrow developer enthusiasm.

Gatekeepers do not operate in a vacuum - they face push and pull forces that shape their technology regulation choices, including:

- Organizational strategies, cultures, incentives and policies
- Existing legal and regulatory frameworks
- Market dynamics and pressures from public groups
- Dominant ethical standards espoused by peers

The interaction between external constraints and individual volition regarding AI releases is a complex dynamic. Our framework acknowledges the impact of contextual factors on governance decisions, accounting for forces beyond personal control. Groups of synthetic gatekeepers, with different hierarchical ranks and expert roles, could simulate maneuverability, leverage, and influence, as a series of nested containment confidences and escape vectors.

Transparent Gatekeepers

We propose Centaur Operant Profiles for synthetic gatekeepers as "Gatekeeper Cards," inspired to compliment existing Model Cards, that use synthetic gatekeepers modeled after developer worldviews and motivations entwined within the sociotechnical environment the AI. Such visibility into the minds of machine makers supports external debate on system soundness. We see candid self-appraisal by inventors as being core to transparency - articulating ethical stances not just for finished models but their underlying human architectures. Their friction with organizations reveals narrow AI oversight concentrated in few fallible hands facing overbearing commercial and competitive pressures. As artificial intelligence capacities accelerate, development and deployment governance depends critically upon appropriate alignment across personal, professional and societal dimensions. The Centaur Box Experiment is a first step towards visualizing this complex multidimensional balancing act between relevant stakeholders necessary for stable co-existence between fallible gatekeepers and exponentially growing technological progeny.

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